

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	15 February 2025
Team ID	EL2026TMID10104
Project Name	AI-Based Anomaly Detection Platform
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a developer, I can gather relevant healthcare data so that the system can analyze patient vitals.	2	High	Rutuja
Sprint-1	Data Collection	USN-2	As a developer, I can load collected data into the system for further processing.	1	High	Rutuja
Sprint-1	Data Preparation	USN-3	As a developer, I can handle missing values in the dataset to improve data quality.	3	High	Rutuja
Sprint-1	Data Preparation	USN-4	As a developer, I can create required fields from raw healthcare data for analysis.	3	Medium	Rutuja
Sprint-1	Data Preparation	USN-5	As a developer, I can remove inconsistencies in the dataset to ensure accurate results.	3	High	Rutuja
Sprint-2	Data Visualization	USN-6	As a user, I can view healthcare data using bar charts to understand comparisons easily.	2	Medium	Rutuja
Sprint-2	Data Visualization	USN-7	As a user, I can view healthcare data using pie charts to understand proportions.	2	Medium	Krupal

Sprint-2	Data Visualization	USN-8	As a user, I can view healthcare data using line charts to analyze trends over time.	2	Medium	Ketan
Sprint-2	Data Visualization	USN-9	As a user, I can visualize healthcare data on a map to understand geographical patterns.	4	Low	Kunal
Sprint-2	Dashboard Development	USN-10	As a user, I can access a dashboard that displays all healthcare insights in one place.	5	High	Ketan
Sprint-2	Story Development	USN-11	As a user, I can view the final analytical story summarizing healthcare insights.	5	Medium	Rutuja

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed	Sprint Release Date (Actual)
Sprint-1	12	6 Days	01 March 2026	06 March 2026	12	06 March 2026
Sprint-2	20	6 Days	08 March 2026	13 March 2026	—	—

Total story points=32

Velocity Calculation

Velocity measures the amount of work a team can complete during a sprint.

Formula

Velocity = Total Story Points Completed / Number of Sprints

Calculation

Total Story Points = **32**

Number of Sprints = **2**

Velocity =

$32 / 2 = \mathbf{16 \text{ Story Points per Sprint}}$

This means the development team can complete **approximately 16 story points in each sprint**, which helps in planning future sprints more accurately.

Average Velocity per Day

If the sprint duration is **10 days** and the team velocity is **20 story points per sprint**, the **average velocity per day** can be calculated as:

Average Velocity (AV) = Story Points / Sprint Duration

AV = 20 / 10

AV = **2 Story Points per Day**

This means the team can complete approximately **2 story points per day**.

Burndown Chart

A **Burndown Chart** is a graphical representation that shows the amount of work remaining versus time during a sprint. It helps teams track progress and determine whether they are on schedule to complete the planned work.

In the burndown chart:

- **X-axis** represents time (days of the sprint)
- **Y-axis** represents remaining story points
- The chart gradually declines as tasks are completed

Burndown charts are widely used in **Agile and Scrum methodologies** to monitor sprint performance and ensure timely delivery of project milestones.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

Reference:

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>